Cybersecurity Command Cheatsheet

A small, safe reference for learning paths: Windows, Linux, logs, networking, and local web labs

NOTEBOOK STYLE Handwriting-inspired typeset notes for printing and revision; not a scan of actual handwriting.

SAFE USE Use these commands on your own device, supplied sample files, and a local training app. Do not test public systems without explicit permission.

1. Windows basics (PowerShell)

Goal	Read-only command / example
See running processes	Get-Process | Sort-Object CPU -Descending | Select-Object -First 10 Name,Id,CPU
Read recent authentication events (permissions may be required)	Get-WinEvent -FilterHashtable @{LogName="Security"; Id=4624,4625} -MaxEvents 20
Read JSON training data	$data = Get-Content .\sample.json -Raw | ConvertFrom-Json
Find a word in a sample file	Select-String -Path .\sample.csv -Pattern "suspicious"

2. Linux basics

Goal	Command / example
Show addresses on your own device	ip addr
List listening sockets on your own device	ss -tulpn
Read recent warning-level journal entries	journalctl -p warning -n 50
Search a local log file for a word	grep -in "failed" sample.log
Count repeated first-column values (CSV must be simple)	cut -d, -f1 sample.csv | sort | uniq -c

3. Networking concepts

Question	Safe example
What address does a public training domain resolve to?	nslookup example.com
What HTTP headers does my local training app return?	curl -I http://127.0.0.1:3000
What is a port?	A numbered endpoint used by a network service; do not assume an open port is a vulnerability.
What is DNS?	A naming system that helps map names to records such as IP addresses.

4. Log triage workflow

1. Record the time range and timezone.
2. Filter the provided synthetic sample to the event types relevant to the question.
3. Group by user, host, source, and outcome where useful.
4. Compare successful and failed activity with a baseline.
5. Keep raw events and note which filters you applied.
6. Write facts, hypotheses, unknowns, and next steps separately.

5. Threat-hunting memory aid

HYPOTHESIS → QUERY → VALIDATE → DOCUMENT What would I expect if my hypothesis were true? What else could explain the same pattern? What data is missing?

6. Bug-bounty checklist

- Read the program brief, exact in-scope assets, exclusions, rate limits, and disclosure rules.
- Prefer official training platforms or a local lab for practice.
- Use minimal, non-destructive tests and demo accounts only.
- Stop if real user data or a production-side effect appears.
- Write a reproducible report and redact secrets, tokens, and personal data.

7. Useful formulas

Metric	Formula	Interpretation
Precision	TP / (TP + FP)	Of the rule alerts, how many were true positives in labelled data?
Recall	TP / (TP + FN)	How many labelled suspicious cases did it find?
Specificity	TN / (TN + FP)	How many labelled benign cases did it ignore correctly?
F1	2 × Precision × Recall / (Precision + Recall)	A combined measure of precision and recall.

A real lab for these notes

Topic: Beginner commands for Linux, Windows CMD, PowerShell, local networking, and safe lab setup. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- Read the help page before using a new command.
- Use read-only discovery first: identity, path, version, and configuration.
- Networking commands below are limited to localhost or a host/IP explicitly assigned by a training room.
- Use git clone only for an intended public repository such as OWASP Juice Shop; the clone command does not start or test any remote target.

Windows Command Line *[TryHackMe]*
https://tryhackme.com/room/windowscommandline
Access note: Account required; a deployable machine is described by the provider. Current room access may vary.

Windows PowerShell *[TryHackMe]*
https://tryhackme.com/room/windowspowershell
Access note: Account required; the provider page may reserve some room tasks for paid access.

Bandit Level 0 *[OverTheWire]*
https://overthewire.org/wargames/bandit/bandit0.html
Access note: Free SSH practice against the official Bandit game host; use only its designated login and port.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

The Nmap examples are scoped to loopback or the exact IP/hostname assigned in the official training room. No blanket permission is implied for public systems.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

LINUX BASICS

```
pwd; ls -la; whoami; uname -a
```

Show directory, files, current user, and OS details.

LINUX LOG READING

```
grep -inE "failed|denied|error" ./sample.log | head -n 20
```

Filter a local training log; adjust keywords to the exercise.

WINDOWS CMD

```
whoami & cd & ipconfig /all
```

Display user, current directory, and local network configuration on your own VM.

POWERSHELL / OS

```
Get-ComputerInfo | Select-Object WindowsProductName,WindowsVersion,OsArchitecture
```

Display OS information; no settings are changed.

POWERSHELL / DNS

```
Resolve-DnsName example.com
```

Demonstrate a simple DNS lookup. Use only queries appropriate to the course and network rules.

NMAP / LOCALHOST ONLY

```
nmap -sT -Pn --top-ports 20 127.0.0.1
```

A basic TCP scan of your own loopback interface. For hosted labs, scan only the IP/hostname assigned by that lab.

NMAP / LAB-ASSIGNED TARGET

```
nmap -sT -Pn -p 22,80,443 "$LAB_TARGET"
```

Replace LAB_TARGET only with the address explicitly provided by the training room; do not substitute a public or third-party address.

OWASP JUICE SHOP / OBTAIN SOURCE

```
git clone --depth 1 https://github.com/juice-shop/juice-shop.git
```

Fetch the official open-source training app source into your current directory. This operation is not a scan.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Choose the right shell: Bash for Linux, cmd.exe for Windows CMD, or PowerShell for Windows scripting.
- 2. Copy one command at a time; check the current directory and target before running it.
- 3. Prefer local sample files and 127.0.0.1 before using a hosted target.
- 4. For scanning, use the exact range written in the course's scope; keep a copy of that scope in your notes.
- 5. Record what a command does and what output would change your next decision.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q What is the difference between `ipconfig` and `nmap`?

Think it through: ipconfig displays the local Windows network configuration; nmap sends probes to the specified target.

Q Can `nmap -sV` be used against any public IP?

Think it through: No. Scan only localhost, your own network with permission, or a target explicitly in scope for the lab.

Q Why is `git clone` not the same as scanning the Git host?

Think it through: It requests repository data from the specified remote; it does not probe the repository host's ports or services.

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.